

Day 3: Introduction to IP Addressing Theory (1 hour): IPv4 and IPv6: Differences, structure, and importance. Subnetting basics: CIDR, subnet masks. Practical (1 hour): Assign static IPs to devices and verify connectivity. Task: Solve subnetting exercises and calculate valid subnets.

Task 1 :- List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

ANS :-

Feature	IPv4	IPv6
Address Length	32 bits (Short: e.g., 192.168.1.1)	128 bits (Long: e.g., 2001:db8::1)
Address Supply	~4.3 billion (Running out)	Virtually endless (Trillions of addresses)
Format	Numbers separated by dots	Hexadecimal numbers separated by colons
Security & Speed	Basic security; needs NAT (address sharing)	Built-in security (IPsec); faster direct connections

Task 2 :- Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

ANS :-

Network Setup:

- Subnet Mask: 255.255.255.0 (same for all)
- Default Gateway: 192.168.10.1 (your router)

1. Laptop

- IP: 192.168.10.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1

2. Phone

- IP: 192.168.10.11
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1

3. Smart TV

- IP: 192.168.10.12
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1

4. Printer

- IP: 192.168.10.13
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1

Task 3 :- Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.

ANS :-

Totals:

- Total Valid Subnets: 8
- Usable Hosts per Subnet: 30 (32 total IPs minus network and broadcast addresses)

First 3 Subnet Ranges:

1. Subnet 1

- Range (Network to Broadcast): 192.168.1.0 to 192.168.1.31
- Usable Host IPs: 192.168.1.1 to 192.168.1.30

2. Subnet 2

- Range (Network to Broadcast): 192.168.1.32 to 192.168.1.63
- Usable Host IPs: 192.168.1.33 to 192.168.1.62

3. Subnet 3

- Range (Network to Broadcast): 192.168.1.64 to 192.168.1.95
- Usable Host IPs: 192.168.1.65 to 192.168.1.94

Task 4 :- You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet.

ANS :-

Network Address: 10.0.5.0

Broadcast Address: 10.0.5.255

Valid Host Range: 10.0.5.1 to 10.0.5.254

Task 5 :- Use an online subnet calculator tool (such as <https://www.subnet-calculator.com/>) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.

ANS :-

The screenshot shows the 'Subnet Calculator' interface. The 'Network Class' is set to 'A'. The 'IP Address' is '10.0.5.17', and the 'Hex IP Address' is '0A.00.05.11'. The 'Subnet Mask' is '255.255.255.0', and the 'Wildcard Mask' is '0.0.0.255'. The 'Subnet Bits' are '16', and the 'Mask Bits' are '24'. The 'Maximum Subnets' is '65536', and the 'Hosts per Subnet' is '254'. The 'Host Address Range' is '10.0.5.1 - 10.0.5.254'. The 'Subnet ID' is '10.0.5.0', and the 'Broadcast Address' is '10.0.5.255'. The 'Subnet Bitmap' is '0nnnnnnnn.ssssssss.ssssssss.hhhhhhhh'.

- **Host Address Range: 10.0.5.1 - 10.0.5.254**
- **Subnet ID (Network): 10.0.5.0**
- **Broadcast Address: 10.0.5.255**

Result : Yes, the manual answer matched the online tool's output perfectly.

